Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	02 October 2026
Team ID	SWTID-2026-5414
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Architecture: Employee data import into ServiceNow using Import Sets and Transform Maps

The spreadsheet is used to import employee information into ServiceNow using Import Sets and Transform Maps. It contains employee details such as Employee ID, Name, Email, Department, and Location. The uploaded spreadsheet data is first stored in an Import Set Table for temporary processing and validation. A Transform Map is then used to map these fields to the corresponding fields in the target Employee table. The system uses Employee ID as the unique identifier to identify existing employees and update their records, while new employees are inserted as new records.

Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

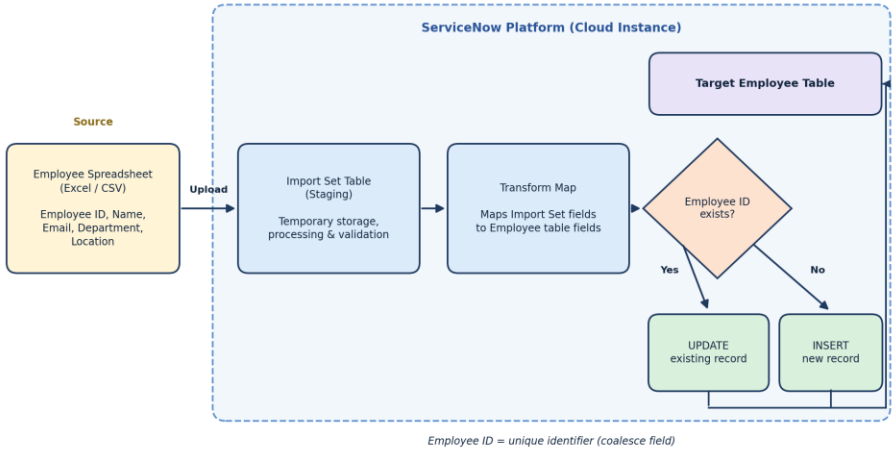


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employee spreadsheet (Employee ID, Name, Email, Department, Location) uploaded by the user through the ServiceNow Import Set / Data Source interface	ServiceNow Platform UI, Excel / CSV
2.	Application Logic-1	Import Set Table: temporary storage of uploaded data for processing and validation	ServiceNow Import Set Table
3.	Application Logic-2	Transform Map: maps Import Set fields to the corresponding fields in the target Employee table	ServiceNow Transform Map
4.	Application Logic-3	Employee ID is the unique identifier: existing employees are updated, new employees are	Transform Map (coalesce on Employee ID)

		inserted as new records	
5.	Database	Target Employee table holding Employee ID, Name, Email, Department and Location	ServiceNow Employee table
6.	Cloud Database	Import Set and Employee tables are stored in the ServiceNow instance database	ServiceNow Cloud Platform
7.	File Storage	Spreadsheet file used as the data source for the import	Excel / CSV file
8.	External API-1	Not applicable	Not applicable
9.	External API-2	Not applicable	Not applicable
10.	Machine Learning Model	Not applicable	Not applicable
11.	Infrastructure (Server / Cloud)	Application runs on the ServiceNow cloud instance Local Server Configuration: Not applicable Cloud Server Configuration: ServiceNow instance	ServiceNow Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Not applicable - implemented using ServiceNow platform features	ServiceNow Import Sets, Transform Maps
2.	Security Implementations	Access to import data and the Employee table is controlled through ServiceNow roles and access controls	ServiceNow Roles, ACLs
3.	Scalable Architecture	Staging in an Import Set Table separates data loading from transformation, supporting bulk	ServiceNow Import Sets, Transform Maps

S.No	Characteristics	Description	Technology
		imports	
4.	Availability	Application is hosted on the ServiceNow cloud platform	ServiceNow Cloud Platform
5.	Performance	Data is validated in the staging table before being loaded into the target Employee table	Import Set Table, Transform Map

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d